

Ayush Maurya

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SUMMARY

Builds deep learning models that deliver measurable results. Trained a ResNet50 geolocation classifier with 92% region accuracy on a custom multi-class dataset. Designed comparative ML experiments evaluating transfer learning against ensemble methods for age prediction. Engineered an OCR-to-LLM NLP pipeline that digitizes historical documents end to end. Published ElectoralSim, an open-source agent-based simulation framework, on PyPI. Currently an M.S. Research candidate at IIIT Hyderabad working across computer vision, NLP, and statistical modeling.

EDUCATION

IIIT Hyderabad B.Tech in Computer Science and M.S. by Research in Computing and Human Sciences Dual degree offered by the Human Sciences Research Center	2021–Present
Mary Gardiner's Convent School Class XII — 90.1%	2021
City Montessori School Class X — 94.2%	2019

EXPERIENCE

SERC Digital Twin • Developed a real-time environmental monitoring dashboard with IoT sensor data pipeline.	<i>Environmental IoT</i>
TAFEA • Integrated transformer-based LLM APIs into a student client platform.	<i>System Design</i>

PROJECTS

ML for Local Geoguesser • Built a ResNet50 deep learning pipeline in PyTorch, achieving 92% region accuracy on geolocation classification.	Deep Learning, Computer Vision
Age Prediction Model • Conducted comparative ML experiments evaluating transfer learning against ensemble methods for age classification accuracy.	Transfer Learning
Highland History • Engineered an NLP data pipeline combining OCR with transformer-based LLMs to digitize historical Himalayan trade documents.	OCR + LLM Pipeline
ElectoralSim • Published an open-source Python library on PyPI for simulating electoral systems with vectorized agents.	PyPI Package
Design for Social Innovation • Built an AI platform with LLM APIs helping Teach For India fellows plan extracurricular curricula.	AI-Assisted Education

TECHNICAL SKILLS

Languages: Python, C/C++, JavaScript, SQL, HTML/CSS
ML/AI: PyTorch, Deep Learning, Computer Vision, NLP, Transformers, LLMs, Classification, Model Evaluation, Feature Engineering, Data Pipeline, Transfer Learning, NumPy, Pandas, Statistical Analysis, Hyperparameter Tuning
Technologies: Linux, System Design, Git

HONOURS

UGEE Rank: 111 out of 40,000+ candidates.

ACTIVITIES

Bass guitarist in college band and lawn tennis player.